The Hong Kong Academy for Gifted Education 香港資優教育學苑

IMO Training 2023/24 Phase II

Inequalities II

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1. A.M. - G.M. Inequality (算術及幾何平均不等式).

For any n (n>1) non-negative real no. $a_1, a_2, a_3, \ldots a_n$,

$$\frac{a_1 + a_2 + a_3 + \ldots + a_n}{n} \geqq (a_1 a_2 a \ \ldots a_n)^{\frac{1}{n}}$$

(or $a_1{}^n + a_2{}^n + a_2{}^n + \cdots + a_n{}^n \geq n\, a_1 a_2 a_3 \ldots a_n$)

The equality holds if and only if $a_1 = a_2 = a_3 = \ldots = a_n$.

<u>Proof using M.I.</u>

For n = 2, $(\sqrt{a} - \sqrt{b})^2 \geq 0$, hence $\frac{a+b}{2} \geq \sqrt{ab}$.

Assume that P is true for any n-1 non-negative real numbers.

That is $\frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1} \geq (a_1 a_2 \ldots a_{n-1})^{\frac{1}{n-1}}$.

Consider any n non-negative real numbers, $a_1, a_2, \ldots, a_n$.

Without loss of generality, assume that $a_1 \leq a_2 \leq \ldots \leq a_n$.

Further, if $a_1 = a_2 = \ldots = a_n$, the inequality holds, otherwise $\frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1} < a_n$,

Now $\frac{1}{n}(a_1 + a_2 + \ldots + a_{n-1} + a_n) = \frac{1}{n}\left[(n-1)\left(\frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1}\right) + a_n\right]$

$$= \frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1} + \frac{a_n - \frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1}}{n}$$

Using the inequality $(x+y)^n \geq x^n + nx^{n-1}y$, we have

$$\left[\frac{1}{n}(a_1 + a_2 + \ldots + a_{n-1} + a_n)\right]^n \geq \left(\frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1}\right)^n + n\left(\frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1}\right)^{n-1}\left(\frac{a_n - \frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1}}{n}\right)$$

$$= a_n \left(\frac{a_1 + a_2 + \cdots + a_{n-1}}{n-1}\right)^{n-1}$$

$$\geq a_n (a_1 a_2 \ldots a_{n-1})$$

$$= a_1 a_2 \ldots a_{n-1} a_n$$

Hence $\frac{1}{n}(a_1 + a_2 + \ldots + a_{n-1} + a_n) \geq (a_1 a_2 \ldots a_{n-1} a_n)^{\frac{1}{n}}$.

[P] is also true when we have n non-negative real numbers..

By the Principle of Mathematical Induction, [P] is true for any positive integer n.

<u>Ex. 1</u>: Let a, b, c be non-negative real numbers. Prove that

$$(a + b + c)^2 \geq 3(a\sqrt{bc} + b\sqrt{ca} + c\sqrt{ab}).$$

<u>Ex. 2</u>: Given that n is a positive integer and a is a real number greater than 1, prove that

$$\frac{a^{n+1} - a^{-(n+1)}}{a^n - a^{-n}} > \frac{n+1}{na}$$

<u>Ex. 3</u>: Let x, y and y be positive real numbers such that x + y + z = 1. Prove that

(i) $x^2yz \leqq \frac{1}{64}$, (ii) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geqq 9$, (iii) $(\frac{1}{x}-1)(\frac{1}{y}-1)(\frac{1}{z}-1) \geqq 8$.

<u>Ex. 4</u>: Prove that if a, b, c are positive real numbers such that a+b+c = 1, then $a^3 + b^3 + c^3 \geq \frac{1}{9}$.

<u>Ex. 5</u>: For any non-negative real numbers a, b and c, prove that $\frac{c}{a} + \frac{a}{b+c} + \frac{b}{c} \geq 2$.

When does the equality hold?

<u>Ex. 6</u>: Given that a>b>c>0, find the minimum of $2a^2 + \frac{1}{ab} + \frac{1}{a(a-b)} - 6ac + 9c^2$.

When does this happen?

Ex. 7: Prove that if a and b are positive real numbers such that a+b = 1, then

$$(a+\frac{1}{2b^2})(b+\frac{1}{2a^2}) \geq \frac{25}{4}.$$

Ex. 8: For real numbers a>1, b>1, prove that $\frac{a^2}{b-1}+\frac{b^2}{a-1} \geq 8$

Ex. 9: Let a, b be real numbers such that a+b = 17. Find the smallest value of $2^a + 4^b$.

Ex. 10: For any positive value of x, find the minimum value of $x^3 - 3x + 5$.

Ex. 11: For any positive integer n > 1, prove that

(a) $2.4.6.....(2n) < (n+1)^n$. (b) $(n!)^3 < n^n (\frac{n+1}{2})^{2n}$

Ex. 12: Prove that $37^{73} > 73!$.

Ex. 13: Prove that for any positive integer, $1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} > n[(1+n)^{\frac{1}{n}} - 1]$

Ex. 14: For any positive integer n, prove that $(1+\frac{1}{n})^n < (1+\frac{1}{n+1})^{n+1} < 3$.

Ex. 15: Given that $a_1, a_2, \ldots, a_n$ are n positive real numbers such that $a_1a_2\ldots a_n = 1$.

Prove that $(2+a_1)(2+a_2)\ldots(2+a_n) \geq 3^n$.

Ex. 16: Let a, b, c, d be positive real numbers such that a+b+c+d = 4. Prove that

$$\frac{a}{b^2+1}+\frac{b}{c^2+1}+\frac{c}{d^2+1}+\frac{d}{a^2+1} \geq 2.$$

Ex. 17: Prove that for any positive integer n,

$$1 < \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{3n+1} < 2.$$

2. Weighted AM-GM Inequality (加權算術及幾何平均不等式)

Let $a_1, a_2, \ldots, a_n$ be non-negative real numbers, and $\phi_1, \phi_2, \cdots, \phi_n$ be non-negative real numbers such that $\phi_1 + \phi_2 + \cdots + \phi_n = 1$, then

$$\phi_1 a_1 + \phi_2 a_2 + \cdots + \phi_n a_n \geq (a_1^{\phi_1})(a_2^{\phi_2}) \ldots\ldots (a_n^{\phi_n})$$

In other words, $\sum_{i=1}^{n} \phi_i a_i \geq \prod_{i=1}^{n} (a_i^{\phi_i})$

Proof for the case n = 2:

Let p and q be non-negative integers and x, y be non-negative real numbers.

Take p 'x's and q'y's, and using A.M. ≥ G.M., $\frac{px+qy}{p+q} \geq (x^p y^q)^{\frac{1}{p+q}}$

Let $\alpha = \frac{p}{p+q}$, and $\beta = \frac{q}{p+q}$, then $\alpha + \beta = 1$

The inequality becomes $\alpha x + \beta y \geq x^{\alpha} y^{\beta}$.

The proof for the general case is similar.

Ex. 18: (62th IMO (2020) No. 2) The real numbers a, b, c, d are such that $a \geq b \geq c \geq d > 0$ and a+b+c+d = 1. Prove that $(a+2b+3c+4d)\, a^a b^b c^c d^d < 1$.

3. Cauchy-Schwarz Inequality (柯西不等式)

(a) For any four non-negative real numbers a, b, c, d, $(a^2 + b^2)(c^2 + d^2) \geq (ac + bd)^2$.

The equality hold if and only if $\frac{a}{c} = \frac{b}{d}$.

(b) Generalizing, for any two sequences of non-negative real numbers $a_1, a_2, a_3, \ldots a_n$ and $b_1, b_2, b_3, \ldots b_n$,

$$[a_1^2 + a_2^2 + a_3^2 + \ldots a_n^2][b_1^2 + b_2^2 + b_3^2 + \ldots b_n^2] \geqq (a_1b_1 + a_2b_2 + a_3b_3 + \ldots + a_nb_n)^2$$

The equality holds if and only if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \ldots = \frac{a_n}{b_n}$.

Proof: (Method 1) $(\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} b_i^2) - (\sum_{i=1}^{n} a_i b_i)^2 = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left(a_i^2 b_j^2 + a_j^2 b_i^2\right) - 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left(a_i b_i a_j b_j\right)$

$$= \sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left(a_i b_j - a_j b_i\right)^2 \geq 0$$

Hence $(\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} b_i^2) \geq (\sum_{i=1}^{n} a_i b_i)^2$

Equality holds if and only if $a_i b_j = a_j b_i$ for all i, j; i.e. $\frac{a_i}{b_i} = \frac{a_j}{b_j}$ for all i, j.

(Method 2): Consider the inequality $\sum_{i=1}^{n}(a_i x - b_i)^2 \geq 0$, i.e. $(\sum_{i=1}^{n} a_i^2)x^2 - 2(\sum_{i=1}^{n} a_i b_i)x + (\sum_{i=1}^{n} b_i^2) \geq 0$

The necessary condition is (i) $(\sum_{i=1}^{n} a_i^2) > 0$ and (ii) $(2\sum_{i=1}^{n} a_i b_i)^2 \leq 4(\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} b_i^2)$

Hence $(\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} b_i^2) \geq (\sum_{i=1}^{n} a_i b_i)^2$

Equality holds only when $a_i x - b_i = 0$ for all i, j; i.e. $\frac{a_i}{b_i} = \frac{a_j}{b_j}$ for all i, j.

Ex. 19: Prove that for any sequence of n real numbers $a_1, a_2, \ldots, a_n$,

(i) $\sum a_i \leq \sqrt{n\sum a_i^2}$

(ii) If all the numbers are positive, then $(\sum a_i)(\sum \frac{1}{a_i}) \geq n^2$.

Ex. 20: Prove that for any real numbers a, b, c, d, $(a+b+c+d)^2 \leq 3(a^2 + b^2 + c^2 + d^2) + 6ab$.

Ex. 21: Let a be a real number, prove that $2\sqrt{a-4} + 3\sqrt{5-a} \leq \sqrt{13}$. When does the equality hold?

Ex. 22: Given that x, y are real numbers satisfying $3x^2 + 2y^2 \leq 6$. Prove that $2x + y \leq \sqrt{11}$.

Ex. 23: Let a, b and c be real numbers such that a + b + c = 1. Find the minimum value of $2a^2 + 3b^2 + c^2$. When does the equality hold?

Ex. 24: If a>0, b>0 and a+b = 1, find the minimum value of $(\frac{1}{a^2} - 1)((\frac{1}{b^2} - 1)$.

Ex. 25: If a, b, c, x, y, z are positive real numbers such that $a^2 + b^2 + c^2 = 36$, $x^2 + y^2 + z^2 = 49$, $ax + by + cz = 42$, find the value of $\frac{a+b+c}{x+y+z}$.

Ex. 26: Given $a > -\frac{1}{2}$ and $b>-1$, and $a+b = 1$, find the maximum value of $\sqrt{2a+1} + \sqrt{b+1}$.

Ex. 27: Let $a_1, a_2, \ldots, a_n$ be n (n>1) positive real numbers, and m be their arithmetic mean. Prove that $nm^2 \leq a_1^2 + a_2^2 + \ldots + a_n^2 < n^2m^2$

Ex. 28: Given that $x_1, x_2, \ldots, x_n$ are n positive numbers, prove that

$$\frac{x_1+x_2+\cdots+x_n+\sqrt{x_1^2+x_2^2+\cdots+x_n^2}}{(x_1^2+x_2^2+\cdots+x_n^2)(\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n})} \leq \frac{n+\sqrt{n}}{n^2}.$$

Ex. 29: Prove that for any positive integer n, $1 + \frac{1}{2} + \frac{1}{3} + \ldots \frac{1}{n} < \sqrt{2n-1}$

Ex. 30: Given that a, b, c, d and e are real numbers such that a + b + c + d + e = 8, and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$. Find the maximum and minimum value of e.

Ex. 31: (36th IMO (1995) No. 2) Let a, b and c be positive real numbers such that abc = 1.

Prove that $\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}$.

4. Harmonic Mean & Root Mean Square (調和平均與均方根)

(a) For any n (n>1) positive real numbers $a_1, a_2, a_3, \dots a_n$, the Harmonic Mean (調和平均)

$H_n = \dfrac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{3_3}+\cdots+\frac{1}{a_n}}$ and the R.M.S.(均方根) is $\sqrt{\dfrac{a_1^2+a_2^2+\dots+a_n^2}{n}}$

Theorem : R.M.S. ≧ A.M. ≧ G.M ≧ H.M.

(i) To prove that H.M. ≤ G.M.

Proof: Since $\left(\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+\dots+\frac{1}{a_n}\right)\left(\frac{1}{n}\right) \geq \left(\frac{1}{a_1a_2a_3\dots a_n}\right)^{\frac{1}{n}}$

Hence $\dfrac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{3_3}+\cdots+\frac{1}{a_n}} \leq (a_1a_2a\ \dots a_n)^{\frac{1}{n}}$

(ii) To proof that R.M.S. ≧ A.M.:

Using Cauchy's Inequality, $(a_1^2 + a_2^2 + \dots + a_n^2)(1+1+\dots+1) \geq (a_1 + a_2 + \dots + a_n)^2$

Hence $n\,(a_1^2 + a_2^2 + \dots + a_n^2) \geq (a_1 + a_2 + \dots + a_n)^2$

It means that $\sqrt{\dfrac{a_1^2+a_2^2+\dots+a_n^2}{n}} \geq \dfrac{a_1+a_2+a_3+\dots+a_n}{n}$

Ex. 32: Let a and b be positive real numbers such that a+b=1. Prove that

(a) $\sqrt{2a+1} + \sqrt{2b+1} \leq 2\sqrt{2}$; (b) $a^4 + b^4 \geq 0.125$.

Ex. 33: Let a, b, c be positive real numbers satisfying $a^2 + b^2 + c^2 = 3$. Prove that

$\frac{1}{1+2ab} + \frac{1}{1+2bc} + \frac{1}{1+2ca} \geq 1$

Ex. 34: If $a_1 + a_2 + a_3 + \dots + a_n = 1+\frac{1}{2n}$, prove that $\sqrt{2a_1+1}+\sqrt{2a_2+1}+\dots+\sqrt{2a_n+1} \leq n + 1$

Ex. 35: Given that x, y are positive real numbers and x+y<27, prove that $\frac{\sqrt{x}+\sqrt{y}}{\sqrt{xy}}+\frac{1}{\sqrt{27-x-y}} \geq 1$'

Ex. 36: For any two sequences of non-negative real numbers $a_1, a_2, a_3, \dots a_n$ and $b_1, b_2, b_3, \dots b_n$, prove that在這裡鍵入方程式。

$$\sqrt{a_1^2+a_2^2+\dots+a_n^2}+\sqrt{b_1^2+b_2^2+\dots+b_n^2} \geqq \sqrt{(a_1+b_1)^2+(a_2+b_2)^2+\dots+(a_n+b_n)^2}$$

When does the inequality hold?

(Note: This is known as the Triangle Inequality (三角不等式))

Case n=1: $|a| + |b| \geqq |a + b|$

Case n=2: $\sqrt{a_1^2+a_2^2} + \sqrt{b_1^2+b_2^2} \geqq \sqrt{(a_1+b_1)^2+(a_2+b_2)^2}$

Case n=3: $[a_1^2 + a_2^2 + a_3^2]^{½} + [b_1^2 + b_2^2 + b_3^2]^{½} \geqq [(a_1 + b_1)^2 + (a_2 + b_2)^2 + (a_3 + b_3)^2]^{½}$

General Case: $\sqrt{\sum_{i=1}^n a_i^2} + \sqrt{\sum_{i=1}^n b_i^2} \geqq \sqrt{\sum_{i=1}^n (a_i+b_i)^2}$

Proof of the General Case: Using Cauchy's Inequality,

$[a_1^2 + a_2^2 + a_3^2 + \dots a_n^2]\,[b_1^2 + b_2^2 + b_3^2 + \dots b_n^2] \geqq (a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n)^2$

$2\sqrt{(a_1^2+a_2^2+\dots+a_n^2)(b_1^2+b_2^2+\dots+b_n^2)} \geqq 2(a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n)$

Hence $\Sigma a_i^2+\Sigma b_i^2 + 2\sqrt{(a_1^2+a_2^2+\dots+a_n^2)(b_1^2+b_2^2+\dots+b_n^2)} \geqq$

$\Sigma a_i^2+\Sigma b_i^2 + 2(a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n)$

$\left(\sqrt{a_1^2+a_2^2+\dots+a_n^2}+\sqrt{b_1^2+b_2^2+\dots+b_n^2}\right)^2 \geqq \Sigma\,[(a_i + b_i)^2]$

$\sqrt{\sum_{i=1}^n a_i^2} + \sqrt{\sum_{i=1}^n b_i^2} \geqq \sqrt{\sum_{i=1}^n (a_i+b_i\}^2}$. Equality holds iff $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n}$.